

Signatures of a Subpopulation of Hierarchical Mergers in the GWTC-4 Gravitational-Wave Dataset

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Repeated black-hole mergers in dense stellar clusters are a plausible mechanism to populate the predicted gap in black hole masses due to the pair-instability supernova process. These hierarchical mergers carry distinct spin characteristics relative to first-generation black holes. We introduce an astrophysically motivated model in the joint space of effective inspiral and precessing spins, which captures the dominant spin dynamics expected for hierarchical mergers. We find decisive evidence for a transition at $m_1 = 46.2^{+12.6}_{-7.2} M_\odot$, above which the population is nearly entirely hierarchical, a location consistent with the anticipated onset of the pair-instability gap. We also infer a global peak in the hierarchical merger rate at $m_1 = 15.7^{+3.2}_{-1.1} M_\odot$. The existence of low- and high-mass subpopulations of higher-generation black holes suggests the contribution of both near-solar-metallicity and metal-poor star clusters to the hierarchical merger population. Our results reinforce the growing evidence for detailed, mass-dependent substructure in the spin distribution of the binary black hole population.

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Introduction—Theoretical models for binary black hole (BBH) formation can be broadly divided into two classes: isolated evolution of field binary stars and dynamical channels (see, e.g., Refs. [1–3] for reviews). Identifying the formation history of an individual merger can be difficult since multiple channels may be consistent with a given set of black-hole parameters. The latest catalog of gravitational-wave (GW) events [4,5] released by the LIGO-Virgo-KAGRA (LVK) Collaboration [6–10] allows for the most detailed investigation yet of the constitution of the BBH population.

It is increasingly clear, both from individual events and population analyses, that massive black holes exist in the range $45 - 120 M_\odot$ [11–14]. This is in tension with the notion of pair-instability supernovae, which stellar evolution theory predicts should leave no black holes in this mass

interval [15–23]. These observations have spurred further investigation into mechanisms that can populate this gap [24–29].

Most observed BBHs are thought to contain two first-generation black holes formed from stellar collapse, creating a second-generation remnant [30,31]. In dense stellar environments, remnants may undergo subsequent coalescences, creating hierarchical mergers: those containing at least one component that is the product of one or more previous mergers [32–37]. Hierarchical mergers are characterized by higher primary masses, isotropic spin orientations [32], and dimensionless spin magnitudes $\chi \approx 0.7$, arising from the orbital angular momenta of their progenitors [38–42], with some spread about $\chi \approx 0.7$ due to their mass ratio and spins [43,44]. These spin predictions are robust across binary star evolution assumptions [45]. Since hierarchical assembly produces more massive black holes, it is a natural candidate for filling the pair-instability supernova (PISN) gap.

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To disentangle hierarchical mergers from first-generation counterparts, it is useful to look in parameters that are both well-measured and in which we expect distinct behavior. The rapid spins, unequal mass ratios, and/or high masses of several observed BBH signals—namely, GW190521 [11], GW231123 [14], and GW241011 [46]—have led to their interpretation as hierarchical mergers [26,47–51], though nonhierarchical interpretations have also been put forward, e.g., [24,52–58]. However, data quality issues and waveform inaccuracies can introduce systematic biases in measurements of individual systems, including their spins [7,14,58–63]. When performing and interpreting population analyses, we therefore turn to *effective parameters*, which are typically more reliably measured than component spin magnitudes and tilts [64–66]. Effective inspiral spin [67,68] is typically the best-measured combination of spin parameters [69], defined as the mass-weighted projection of the spin vectors onto the orbital angular momentum: $\chi_{\text{eff}} = (\chi_1 \cos \tau_1 + q\chi_2 \cos \tau_2)/(1+q)$, where q denotes the mass ratio (defined on $0 < q \leq 1$), χ_1 and χ_2 are the respective spin magnitudes of the primary (heavier) and secondary (lighter) black holes, and $\tau_{1,2}$ are the angles between the spin vectors and the Newtonian orbital angular momentum at a chosen reference frequency.

The natal spins of black holes are theoretically uncertain but often expected to be small [70–74], in which case first-generation BBHs (1G + 1G) will have a narrow χ_{eff} distribution centered near zero. Isotropically oriented dynamical mergers create a distribution symmetric around zero, while preferentially aligned isolated binaries would induce a positive centroid [68,75]. In contrast, hierarchical mergers between one first-generation and one second-generation black hole (1G + 2G) will have a uniform χ_{eff} distribution centered at 0 with a width of ~ 1 [25]. As higher-generation merger rates are likely an order of magnitude below that of 1G + 2G mergers [76–78], we specialize to the latter.

As spin parameters are useful differentiators between formation channels, many studies have searched for population-level signatures in χ_{eff} , spin magnitudes, and/or spin orientations, often in conjunction with mass dependence [79–102]. Recently, Refs. [93,94,99] found indication of a transition to higher spin magnitudes in the population at primary masses above $m_1 \sim 40 - 50 M_\odot$, and Refs. [95–98,100] presented strong evidence for a transition in χ_{eff} from narrow to broad above $m_1 \sim 45 M_\odot$. References [97,100] furthermore demonstrated the χ_{eff} transition to be coincident with a truncation in the secondary mass distribution, remarkably in line with the theoretical expectation for the lower edge of the PISN gap. These studies suggest that at $m_1 \gtrsim 45 M_\odot$ the population is dominated by second-generation black holes.

While effective spin has been a successful population-level diagnostic of hierarchical mergers, it is a lossy compression of the six-dimensional spin space to a single

parameter. Hierarchical mergers are distinguished from 1G + 1G mergers not only in χ_{eff} , but cleanly in the *joint* space of χ_{eff} and effective precessing spin χ_p , which is a linearization of the in-plane spin dynamics [103–105]: $\chi_p = \max[\chi_1 \sin \tau_1, (3 + 4q)/(4 + 3q)q\chi_2 \sin \tau_2]$. The joint $\chi_{\text{eff}}\text{--}\chi_p$ distribution for 1G + 1G mergers is expected to cluster near $\chi_{\text{eff}}, \chi_p \sim 0$ [106], while that of 1G + 2G mergers scatters around the boundary of an ellipse centered on 0 with major axis $\chi_p \approx 0.7$ and minor axis $\chi_{\text{eff}} \approx 0.47$ [51,95].

In this Letter, we introduce an astrophysically motivated population model in the joint space of effective inspiral and precessing spin. We include a two-dimensional ellipse-boundary component to capture 1G + 2G mergers alongside an uncorrelated Gaussian to describe 1G + 1G binaries. We explore the mixing fraction between these components as a function of mass and recover strong evidence for a subpopulation of hierarchical mergers at $m_{\text{peak}} = 15.7^{+3.2}_{-1.1} M_\odot$ and a transition at $m_* = 46.2^{+12.6}_{-7.2} M_\odot$ above which the majority of mergers are hierarchical (here and elsewhere we quote the medians and 90% symmetric credible intervals). Our results are consistent with and bolster studies that model only in χ_{eff} by capturing the astrophysical expectation with higher fidelity in the multi-dimensional spin space. We reveal correlated structure in the χ_p distribution that increases credence both for the robustness of spin-mass features and their association with hierarchical mergers.

Models—We analyze BBHs in the Fourth Gravitational-Wave Transient Catalog (GWTC-4) [4] with false-alarm rates $< 1 \text{ yr}^{-1}$ alongside GW241011 and GW241110, two events from the second part of the fourth observing run (O4b) whose spin properties imply a hierarchical origin [46]. We model the population $\chi_{\text{eff}}\text{--}\chi_p$ distribution as a mixture of two components: independent truncated Gaussians for the 1G + 1G distribution and a probability density concentrated on an ellipse boundary for the 1G + 2G distribution. (Throughout, we associate the Gaussian and ellipse-boundary components to 1G + 1G and 1G + 2G mergers, respectively, and refer to the components with their associated astrophysical interpretations interchangeably.) Figure 1 schematically depicts the model components. The Supplemental Material [107] contains further details of our models and hierarchical inference settings, including sensitivity injections and waveform models for single-event parameter estimation.

When secondary spin is negligible, $\chi_{\text{eff}} \approx \chi_1/(1+q) \cos \tau_1$ and $\chi_p \approx \chi_1 \sin \tau_1$. Second-generation black holes have a characteristic spin $\chi_c \sim 0.7$ and typical mass ratios $q_c \sim 0.5$ with their secondary. With isotropic tilts, the astrophysical distribution of 1G + 2G mergers in $\chi_{\text{eff}}\text{--}\chi_p$ follows the boundary of an ellipse with major axis $\chi_p = \chi_c \approx 0.7$, minor axis $\chi_{\text{eff}} = \chi_c/(1+q_c) \approx 0.47$, uniform marginal χ_{eff} , and marginal χ_p declining by

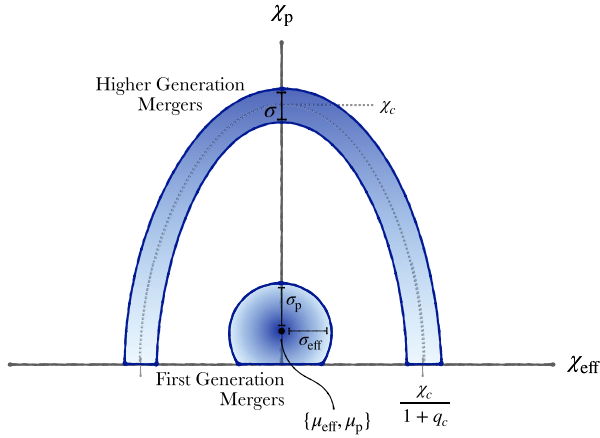


FIG. 1. Schematic of our two-component model in $\chi_{\text{eff}}\text{--}\chi_p$ space. The truncated Gaussian describes 1G + 1G mergers and is defined by its means ($\mu_{\text{eff,p}}$) and standard deviations ($\sigma_{\text{eff,p}}$) in both dimensions. The ellipse-boundary component, ascribed to 1G + 2G mergers, is determined by a characteristic spin χ_c , characteristic mass ratio q_c , and Gaussian width perpendicular to the curve σ .

$p(\chi_p) \propto \chi_p / (\chi_c^2 - \chi_p^2)^{1/2}$ [25,51,95]. The separation in $\chi_{\text{eff}}\text{--}\chi_p$ between first-generation and hierarchical mergers, and the latter's elliptical structure, is pictured in Fig. 1 of Ref. [106] for simulated dynamical mergers from the Cluster Monte Carlo catalog [76,77].

We model the probability density for 1G + 2G mergers as a Gaussian in the perpendicular distance from an ellipse boundary with a filter along χ_p to account for the declining density. The characteristic spin χ_c , characteristic mass ratio q_c , and Gaussian width $\ln \sigma$ are inferred from the data. The χ_{eff} and χ_p distributions of the 1G + 1G component are modeled as uncorrelated Gaussians, though we consider other models in Supplemental Material [107]. We allow the mixing fraction between the Gaussian (1G + 1G) and ellipse-boundary (1G + 2G) components to vary with primary mass, similar to the models employed in Refs. [98,100]. We use a cubic spline to flexibly model the branching ratio without imposing strong priors on the existence or location of transitions; Supplemental Material [107] contains specifics of our node choices. While this is a strong model, we do not enforce the 1G + 2G component's existence; further, it is built on simple, robust astrophysical grounds.

For primary mass, we make the simplifying assumption that the population is described by a single distribution, for which we use the default model of the LVK for GWTC-4: a broken power law with two Gaussian peaks and low-mass smoothing [5]. While we employ the standard power-law mass ratio model for the 1G + 1G component, for the 1G + 2G mergers we instead use a truncated normal, as hierarchical mergers may have a mass ratio peak away from unity. Evolution over cosmic time for both components is modeled with a single power law.

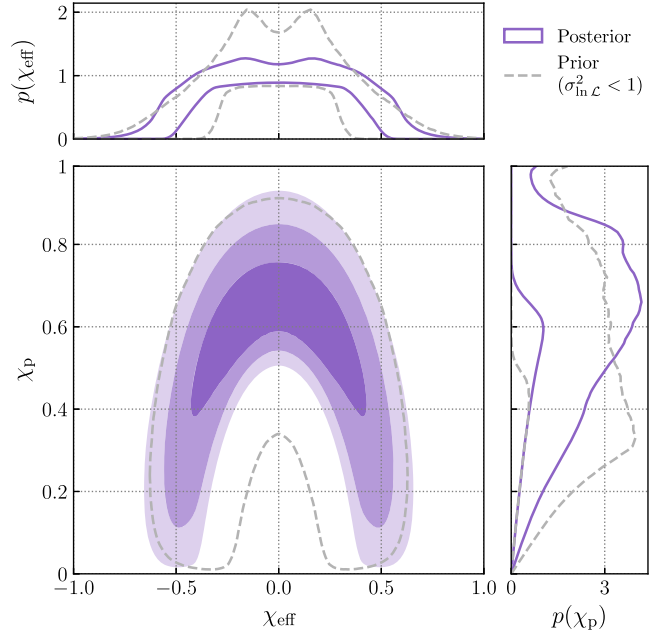


FIG. 2. Posterior on the ellipse-boundary component in $\chi_{\text{eff}}\text{--}\chi_p$ space. The joint distribution plots the posterior median with 50%, 90%, and 99% percentile contours. In dashed gray, we show the 99% credible contour of the median prior draw that passes the likelihood variance threshold. Side panels plot the 90% credible regions of the marginal posteriors and priors.

Results—Figure 2 shows the posterior on χ_{eff} and χ_p for the ellipse-boundary (1G + 2G) component; Supplemental Material [107] Fig. 3 shows the full distribution with both components. The main panel shows the median joint distribution while the marginals show the 90% credible regions of the posterior. Prior draws that pass the likelihood variance cut are in gray. The characteristic spin is inferred to be $\chi_c = 0.66^{+0.18}_{-0.28}$ in agreement with the expectation of $\chi_c \approx 0.7$ for 1G + 2G mergers [42]. Figure 2 illustrates the normalized density of the hierarchical component independent of its population fraction.

Figure 3 plots the fraction of 1G + 2G mergers, ξ , as a function of primary mass. Near $m_1 \sim 46M_\odot$, the 1G + 2G fraction transitions from $\xi \approx 0$ to $\xi > 0$, consistent with Refs. [93–98,100]. While the flexible model does not force a transition, we can use the inferred $\xi(m_1)$ structure to define a transition point m_* as the mass in $m_1 \in (30, 80)M_\odot$ at which ξ first crosses 0.5. We find $m_* = 46.2^{+12.6}_{-7.2}M_\odot$. The 1G + 2G fraction is confidently constrained to be nonzero above this transition—and is consistent with unity—until $\sim 100M_\odot$. Quantitatively, we obtain $\xi > 0.5$ for $m_1 \in [60, 100]M_\odot$ at 90% credibility. Outside $\sim 7\text{--}100M_\odot$, we roughly recover the prior. Marginalized over primary mass, the total fraction of hierarchical mergers is $\xi = 0.14^{+0.12}_{-0.07}$.

In addition to a high-mass population of hierarchical mergers, we observe evidence for a distinct subpopulation of 1G + 2G mergers with masses $m_1 \in (13, 20)M_\odot$. The

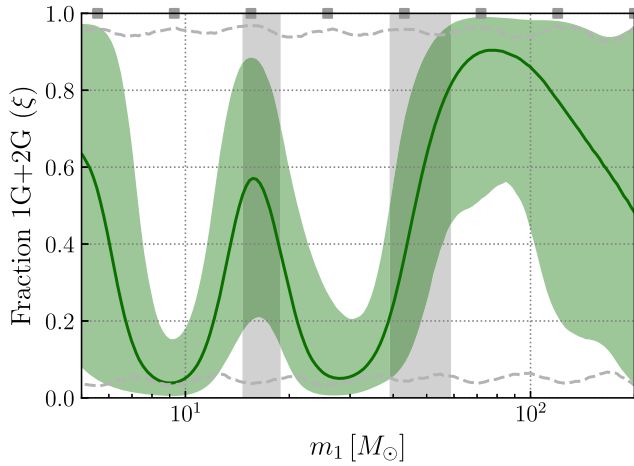


FIG. 3. The inferred fraction of hierarchical mergers $\xi(m_1)$. We plot the median and 90% symmetric credible interval. Gray squares mark the locations of the spline nodes. The dashed lines enclose the 90% region of the prior. The gray vertical regions span the 90% credible regions for the identified low-mass peak m_{peak} and high-mass transition m_* .

1G + 2G fraction is constrained to be $\gtrsim 5\%$ at the 99% level in this range. The existence of this low-mass population is compatible with Refs. [98,100] even as we expand the dimensionality and accuracy of the population model by including spin precession. Taking the value of m_1 that maximizes $\xi(m_1)$ in this mass range, we find a peak in the fraction of hierarchical mergers at $m_{\text{peak}} = 15.7^{+3.2}_{-1.1} M_\odot$. Inference of this feature is not solely driven by the hierarchical candidates GW241011 and GW241110, which lie in this mass regime: an analysis excluding these events also recovers the lower peak (see Supplemental Material [107]), indicating the feature is not an artifact of their inclusion. Including these events without the entirety of O4b introduces a degree of selection bias, though we note this choice is shared by other recent works [100–102], as GWTC-4 alongside these events comprise the most complete data product currently available. While the data suggest potentially a large fraction of mergers in $13 - 20 M_\odot$ are hierarchical, they also imply few mergers in $20 - 40 M_\odot$ are the $\chi_{\text{eff}} - \chi_p$ distribution inferred to be primarily Gaussian (i.e., first-generation) for $m_1 \sim 8 - 10 M_\odot$ and $m_1 \sim 20 - 40 M_\odot$. Our tightest constraints on ξ are in the regimes where we infer a low proportion of hierarchical mergers. At $m_1 = 9 M_\odot (30 M_\odot)$, we constrain $\xi < 0.12 (\xi < 0.16)$ at 90% credibility.

The astrophysically informed ellipse-boundary $\chi_{\text{eff}} - \chi_p$ model for the hierarchical component induces, by design, a quasiuniform marginal χ_{eff} akin to the one-dimensional model employed by Refs. [98,100]. Hence, our comparable identification of spin-mass features is not unexpected. The inclusion of χ_p does not significantly improve constraints (see Supplemental Material [107]), but crucially corroborates the hierarchical hypothesis in the precessing spin

dimension. Further, χ_p enables richer inference of the population spin structure, such as the inferred spin magnitude peak at $\chi \approx 0.7$, which strengthens the emerging picture of a high-spin, isotropic subpopulation consistent with hierarchical assembly.

The inclusion of the 1G + 2G population significantly impacts the inferred spin distribution of the bulk BBH population (also see Supplemental Material [107] Fig. 3). The width of the 1G + 1G Gaussian is, foreseeably, narrower than when the hierarchical component is excluded. In χ_{eff} , the two-component model recovers $\sigma_{\chi_{\text{eff}}} = 0.07^{+0.02}_{-0.01}$, nearly half that of the Gaussian-only model, which returns $\sigma_{\chi_{\text{eff}}} = 0.13^{+0.03}_{-0.02}$. The χ_{eff} mean remains constrained positive, $\mu_{\chi_{\text{eff}}} = 0.03^{+0.02}_{-0.02}$. Neglecting the 1G + 2G subpopulation biases first-generation spin measurements as a one-component model artificially broadens to absorb the hierarchical spread.

The model including the hierarchical component, with the branching ratio an evolving function of mass, is preferred over a pure Gaussian model with a natural log Bayes factor of $\ln \mathcal{B} = 14.5$, which accounts for model complexity. It is favored over a model where the branching ratio is constant over mass with $\ln \mathcal{B} = 6.8$. These Bayes factors indicate significant evidence for structure in the effective inspiral and precessing spin distribution consistent with hierarchical mergers, which, moreover, is nonlinearly correlated with primary mass.

Discussion and conclusion—Using an astrophysically informed mixture model in the joint $\chi_{\text{eff}} - \chi_p$ space, we find strong evidence for a subpopulation of binary black hole mergers that occupy the boundary of an ellipse with a characteristic spin near $\chi_c \sim 0.7$, consistent with the expectation for hierarchical mergers. Allowing the mixing fraction between the components to vary with primary mass, we identify a transition to primarily high-spin and quasiuniform χ_{eff} at $m_* = 46.2^{+12.6}_{-7.2} M_\odot$. Interpreting the elliptical arc component as hierarchical mergers, the data indicate most black holes above $46 M_\odot$ arise from repeated mergers in dense stellar clusters, aligned with the predicted onset of the PISN gap.

Moreover, we measure a peak in the fraction of hierarchical mergers at $m_{\text{peak}} = 15.7^{+3.2}_{-1.1} M_\odot$. Hierarchical assembly is frequently discussed as a method to produce black holes in and above the PISN gap. Indeed, the transition near $46 M_\odot$ corroborates this interpretation. However, simulations of dense stellar clusters suggest the 2G mass distribution declines with primary mass and peaks near $15 M_\odot$, driven by an abundance of low-mass first-generation black holes produced by solar-metallicity stars [34,76,78,120]. As such, the observation of this lower peak in BBH data is unsurprising and may constitute the maximum of the 2G mass spectrum. At the same time, 2G black holes above $46 M_\odot$ are uncommon in near-solar-metallicity clusters, potentially implying the high-mass

component arises from clusters with different initial conditions—namely, lower metallicities—than the low-mass population. Coupling the features in the data with predictions from stellar evolution may enable constraints on the metallicity distribution of the progenitor stars of black-hole mergers.

While the low-mass peak meshes with theoretical expectation, the ostensible *lack* of hierarchical mergers between 20 and $40M_{\odot}$ is perhaps more surprising. The dominance of hierarchical mergers in the $10 - 20M_{\odot}$ range followed by the dearth in $20 - 40M_{\odot}$ may imply either a dip in the first-generation mass spectrum in the former range, or a pileup in the latter. A drop in progenitor core compactness may drive a dip in this lower regime [121], for which several studies have searched and some have tentatively identified [122–124]. If the observed feature at $\sim 35M_{\odot}$ primarily contains first-generation mergers, it may also contribute to the apparent lack of second-generation black holes at moderate masses. Untangling what the inferred branching fraction across primary mass indicates for both the first- and second-generation mass spectra—and the properties of their stellar progenitors—is a key avenue for future study.

Although past work has yielded similar results, no studies have thus far considered the influence of precessing degrees of freedom. Neglecting to model a parameter (which, in practice, means assuming it follows a fixed distribution—here, the single-event parameter estimation prior) can be a source of systematic bias. While we find χ_p does not substantially further constrain the mass-dependent fraction of hierarchical mergers obtained with χ_{eff} only, the consistency between results bolsters confidence in the hierarchical interpretation of effective spin structure while extending into a deeper observable space.

Taken at face value, the fraction of hierarchical mergers inferred from GW data is consistent with a population of entirely dynamical mergers. This meshes with Ref. [125], who note consistency between the BBH merger rate found by the LVK and from cluster simulations. We find the total $1G + 2G$ fraction to be $\xi \sim 14\%$, while $1G + 2G$ mergers comprise $\sim 17\%$ of mergers in the Cluster Monte Carlo catalog [76,77], and the astrophysical hierarchical fraction is likely closer to $\sim 10\%$ [120]. The $1G + 1G \chi_{\text{eff}}$ distribution has support at negative values, which may signify a large contribution of $1G$ dynamical mergers, as expected given the apparent observation of $2G$ dynamical mergers. However, its confidently positive centroid is instead indicative of preferentially aligned isolated binaries. We caution against overinterpreting the similarity in population fraction, as both rely on uncertain assumptions: the former on associating the ellipse-boundary component to $1G + 2G$ mergers, and the latter on prescriptions that control the merger rate such as the escape velocity distribution.

Sequential mergers may also occur in active galactic nuclei disks, preferentially creating aligned mergers

[26,27,126,127]. A symmetric ellipse arc is favored (see Supplemental Material [107]), suggesting active-galactic-nuclei-driven mergers are not dominant constituents of the inferred $\chi_{\text{eff}}-\chi_p$ distribution. Further observations might allow us to elucidate hierarchical subpopulations corresponding to different mechanisms that prompt the repeated mergers.

Linking the black-hole mass spectrum to features in the spin distribution is a promising avenue to more precisely identify hallmarks of stellar evolution that may otherwise be washed out, like the PISN mass gap, by combining information across multiple dimensions. Improved constraints on the PISN gap will sharpen bounds on nuclear processes inaccessible at laboratory temperatures, in particular the cross section of the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ reaction, pivotal to energy production in massive stars [22,128,129]. As the catalog of binary black hole mergers continues to grow alongside advancements in stellar theory and simulation, the statistical significance and associated astrophysical interpretation of spin-mass features in the population will be tested with increasing rigor, emphasizing the power of astrophysically informed population studies to uncover the evolutionary history of black holes.

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Data availability—The data that support the findings of this article are openly available [130].

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